

Human Computer Interaction

Time: 3 Hrs.

Answer All Questions

Question 1

1. Define Human–Computer Interaction (HCI)
2. What is meant by interaction as communication in HCI?
3. What is meant by human perception in the context of HCI?
4. Explain the concept of Human–Computer Interaction and discuss why it is important in modern computing environments.

(Marks 4*5 = 20)

Question 2

1. Explain the main goals of Human–Computer Interaction (HCI)
2. Discuss how poor HCI design can increase support and development costs. Use real-world examples to justify your answer.
3. A hospital Intensive Care Unit (ICU) uses a computerized system to monitor patient vital signs and manage critical medical data. However, the system occasionally crashes and sometimes displays incorrect information. Medical staff depend heavily on this system to make quick decisions, and such failures can place patients and staff at serious risk.
 - a. Explain two reasons why this issue is critical in an ICU environment.
 - b. Suggest two HCI design improvements to address the problem.

(Marks 5*4 = 20)

Question 3

1. Describe the four stages of the User-Centered Design process. **(Marks 5)**
2. Evaluate the advantages and disadvantages of the User-Centered Design approach. **(Marks 5)**
3. You are redesigning an interface for elderly users. Using the human Input–Output channel model (vision, hearing, touch), analyze what sensory limitations must be considered and propose improvements to enhance usability.

(Marks 10)

Question 4

1. Explain Abowd and Beale's Interaction Framework. **(Marks 5)**
2. Briefly Explain Donald Norman's Model of Interaction, explaining both the Gulf of Execution and the Gulf of Evaluation. **(Marks 5)**

3. A user attempts to submit an online form but cannot locate the "Submit" button easily. Using Norman's interaction concepts, explain what might be causing the difficulty and how the Gulf of Execution can be reduced. **(Marks 10)**

Question 5

1. Define Interaction Design
2. Explain the concept of usability in HCI and discuss how it supports effective system development.
3. State Norman's principle "Make things visible" with an example.
4. Write two advantages of using heuristics in HCI.
5. What is the purpose of error prevention in interface design? Give one example.

(Marks 5*4 = 20)